The Incompleteness of Observation

Why Quantum Mechanics and General Relativity Cannot Be Unified From Within

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March 2026

Draft Pre-Print — Theoretical Physics / Foundations

Abstract

Physical observers are strictly subsystems of the deterministic universes they measure, permanently barred from accessing degrees of freedom beyond their causal reach. This paper proves that the structural constraints imposed by this epistemic reduction necessarily give rise to quantum mechanics. We show that an observer coupled to a slow, high-capacity hidden sector on a finite-dimensional configuration space must describe the visible sector using P-indivisible stochastic dynamics. By the stochastic-quantum correspondence, this reduction is mathematically equivalent to unitarily evolving quantum mechanics.

Crucially, the converse also holds: any quantum system, realized as a deterministic dilation, requires precisely these conditions — non-trivial coupling, slow-bath memory, and sufficient hidden-sector capacity — establishing a full equivalence between quantum mechanics and embedded observation. Consequently, the Schrödinger equation, the Born rule, and Bell inequality violations emerge as structural consequences of this equivalence, requiring no independent quantum postulates.

Applying this theorem to the cosmological horizon, which provides an exact physical realization of these causal conditions, uniquely determines the emergent action scale. Partition-relativity, dimensional analysis, and the thermodynamic consistency of the horizon fix $\hbar = c^3\epsilon^2/(4G)$, establishing the geometric discreteness scale at $\epsilon = 2\,l_p$ and recovering the exact Bekenstein-Hawking entropy formula. Furthermore, this framework reframes the cosmological constant problem from a theoretical failure into corroborating evidence. The 10^{122} discrepancy is precisely S_{dS}, representing the exact information compression ratio of the emergent quantum description. Consequently, the observed macroscopic vacuum energy is not a fine-tuned anomaly, but the mandatory classical baseline predicted by the framework. Falsifiable predictions include dark energy evolution in Running Vacuum Model form ($\Lambda_{\mathrm{eff}} = \Lambda_0 + \nu H^2$) with a coefficient $\nu_{\mathrm{OI}} = \Omega_\Lambda/(2\,\ln(c/\epsilon H_0)) \approx 2.45 \times 10^{-3}$, consistent with current observational fits, and distinct gravitational wave echoes near black hole horizons.

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PART I: THE GENERAL THEOREM

1. INTRODUCTION

1.1 The Problem of Embedded Observation

Physical observers are subsystems of the systems they measure. This elementary fact has consequences that have not been traced. An observer embedded in a deterministic universe cannot access the complete state: degrees of freedom beyond the observer's causal reach are permanently hidden, and the observer's description of the visible sector is necessarily a reduced one obtained by marginalizing over the hidden sector. The question is what structural constraints that reduction imposes on the form of the observer's physical laws.

The observer's epistemic situation has been foregrounded before. QBism treats quantum states as expressions of an agent's beliefs; relational quantum mechanics defines states relative to interacting subsystems; 't Hooft's cellular automaton interpretation [1] derives quantum behavior from deterministic dynamics through information loss. These programs take observer-dependence as an interpretive starting point or, in 't Hooft's case, derive it from a specific microphysical model. The present framework differs in that it identifies *necessary and sufficient conditions* under which any embedded observer in any deterministic system necessarily sees quantum mechanics, independent of the system's specific physical content — and proves that these are the *only* conditions under which quantum mechanics arises from a deterministic embedding.

The structural constraints imposed by this reduction are not merely interpretive; they carry massive empirical weight. By establishing this equivalence, the framework reveals that phenomena previously viewed as catastrophic theoretical anomalies—most notably the 10^{122} cosmological constant discrepancy—are in fact mandatory, predictable consequences of embedded observation.

1.2 Axioms

The framework rests on four axioms. None reference quantum mechanics, general relativity, or any specific physical theory.

1. **Deterministic dynamics.** The total system evolves deterministically on a phase space Γ. The phase-space density $\rho(q, p, t)$ evolves via the Liouville equation:

$$\frac{\partial \rho}{\partial t} = \{H_{\text{tot}}, \rho\} \equiv \mathcal{L}\rho$$

where $\mathcal{L}$ is the Liouville operator and $\{\cdot, \cdot\}$ denotes the Poisson bracket.

2. **Finiteness.** The effective configuration space of the visible sector is finite-dimensional, with a discreteness scale ϵ providing a finite minimal cell volume.

Any observer bounded by a finite-area surface can couple to only finitely many modes across that boundary, and holographic arguments independently require the Hilbert space

of quantum gravity to be locally finite-dimensional [27]. These physical constraints motivate the axiom. It enters the proof at a single point — the stochastic-quantum correspondence (§3.1), which requires a finite configuration space. In Part I, ϵ is a free parameter; its value is determined *a posteriori* by self-consistency (§6), which pins $\epsilon = 2\,l_p$ exactly. In Part II, the cosmological horizon provides the finite-area boundary that makes the condition a consequence of the partition geometry rather than an independent assumption.

3. **Causal partition.** Γ is partitioned into a visible sector Γ_V accessible to the observer and a hidden sector Γ_H that is not:

$$\Gamma = \Gamma_V \times \Gamma_H$$

The partition is fixed on timescales relevant to the observer (it may drift on much longer timescales, but such drift is not required for the theorem). The total Hamiltonian decomposes as:

$$H_{\text{tot}} = H_V + H_H + H_{\text{int}}$$

The product decomposition $\Gamma_V \times \Gamma_H$ is a simplification. A real observer's epistemic access is graded: some degrees of freedom are measured precisely, others poorly, and others not at all. The axiom requires that this graded access be well-approximated by a sharp partition — that there exist a clean division into accessible and inaccessible sectors such that cross-partition correlations enter only through H_{int}. This is a condition on the physics, not just the observer: it is satisfied when a physical boundary (causal, thermodynamic, or spatial) enforces a separation sharp enough that the residual leakage across the boundary can be absorbed into the interaction term. The theorem applies to any system where such an approximation holds; Part II identifies a physical realization where the sharpness is exact.

4. **Classical probability.** The observer's predictions are classical expectation values over the phase space; standard results of probability theory (the law of large numbers, the central limit theorem) are the primary statistical tools.

These axioms contain no quantum postulates — no Hilbert spaces, no commutation relations, no Born rule. The framework's central claim is that quantum mechanics *emerges* from these premises under the conditions specified below (§1.3).

1.3 Conditions on the Hidden Sector

The theorem requires three conditions on the relationship between the visible and hidden sectors. These are stated as sufficient conditions for the emergence of quantum mechanics; §3.3 proves they are also necessary, establishing a full equivalence.

(C1) Non-zero coupling. The interaction Hamiltonian satisfies $H_{\text{int}} \neq 0$. The visible and hidden sectors are dynamically coupled, not merely co-existing. The coupling is bidirectional: visible-sector transitions influence the hidden-sector state, and the hidden-sector state influences visible-sector transition probabilities.

(C2) Slow-bath timescale separation. The hidden sector's correlation time τ_B vastly exceeds the visible sector's dynamical timescale τ_S:

$$\tau_S \ll \tau_B$$

This is the *inverse* of the standard Markovian regime, which requires $\tau_B \ll \tau_S$. The hidden sector evolves on timescales far exceeding those of any process accessible to the observer.

(C3) Sufficient capacity. The hidden sector has enough degrees of freedom that visible-sector interactions do not appreciably perturb its state on timescales $\ll \tau_B$. Precisely: if N_H and N_V denote the number of independent degrees of freedom in the hidden and visible sectors respectively, then N_H/N_V is large enough that the cumulative imprint of all visible-sector transitions during any observationally relevant interval occupies a negligible fraction of the hidden sector's state space.

Statement of the theorem. Under Axioms 1–4 and conditions (C1)–(C3), the embedded observer's reduced description of the visible sector is mathematically equivalent to unitarily evolving quantum mechanics. The remainder of Part I constitutes the proof. The conditions are shown to be not merely sufficient but *necessary* for P-indivisibility in any deterministic dilation (§3.3), establishing a full equivalence between quantum mechanics and embedded observation.

1.4 Partition-Relativity

Before proceeding to the proof, a structural consequence of the axioms is derived that will constrain the emergent description throughout.

Lemma (partition-relativity). *The emergent description of the visible sector is uniquely determined by the partition. Two observers whose causal access defines the same partition obtain the same reduced description. Any parameters of the emergent theory can depend only on quantities intrinsic to the partition — the geometric and thermodynamic properties of the boundary between visible and hidden sectors.*

Proof. By Axiom 1, the dynamics are deterministic: the Hamiltonian flow ϕ_t on $\Gamma_V \times \Gamma_H$ is a single, uniquely defined map. By Axiom 3, the partition $\Gamma = \Gamma_V \times \Gamma_H$ is fixed. By Axiom 4, the observer uses the Liouville measure μ on Γ_H — the unique absolutely continuous measure preserved by the Hamiltonian flow (Liouville's theorem); any other smooth measure would evolve under the dynamics, introducing dependence on an arbitrary reference time, and singular invariant measures (e.g., those supported on periodic orbits) are excluded by Axiom 4's requirement that the observer assigns classical probability distributions over the full phase space. The marginalized transition probabilities

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2-t_1}(x_i, h))]\, d\mu(h)$$

are therefore uniquely determined by three inputs: the dynamics (ϕ_t), the partition (Γ_V, Γ_H), and the measure (μ). Since ϕ_t and μ are fixed by the axioms, the transition probabilities — and hence the complete stochastic process — depend only on the partition. If this process is P-indivisible (as shown in §2), the stochastic-quantum correspondence maps it to a quantum description. Since the stochastic process is unique, the quantum description is unique (up to unitary equivalence and gauge freedoms resolved

below). In particular, any free parameters of the emergent theory — including the action scale $\hbar$ — can depend only on quantities intrinsic to the partition: the geometric and thermodynamic properties of the boundary between Γ_V and Γ_H. □

2. EMERGENT STOCHASTICITY AND P-INDIVISIBILITY

2.1 Emergent Stochasticity

The total system (visible + hidden sectors) evolves deterministically (Axiom 1). Every phase-space point $(x, h) \in \Gamma_V \times \Gamma_H$ maps to exactly one future point under the Hamiltonian flow. No randomness exists at this level. But the visible sector alone — what the embedded observer accesses — is stochastic. The observer knows x but not h. Different hidden-sector states h_k compatible with the same x send x to different visible-sector futures x'_k. The observer must assign transition probabilities $T_{ij}(t_2, t_1)$ by marginalizing the deterministic flow over the hidden sector using the Liouville measure, as defined in §1.4. As established there, the Liouville measure is the unique absolutely continuous measure preserved by the Hamiltonian flow, ensuring that the transition probabilities — and hence the emergent description — are uniquely determined by the partition.

This architecture makes the framework a hidden variable theory: the randomness observed by the embedded observer is epistemic, arising from ignorance of hidden-sector degrees of freedom rather than fundamental indeterminacy. The consistency of this structure with Bell's theorem is addressed in §3.2.

2.2 The Slow-Bath Regime

The Markovian limit for reduced dynamics requires a strict timescale separation $\tau_B \ll \tau_S$ [4], where the environment decorrelates much faster than the system evolves. Condition (C2) inverts this hierarchy: $\tau_S \ll \tau_B$. The hidden sector evolves on timescales far exceeding those of any visible-sector process, so its correlations do not decorrelate between successive system transitions. The memory kernel of the reduced dynamics is effectively constant over all observationally relevant timescales.

A slow bath must be distinguished from a static external field. A genuinely static (decoupled) hidden sector would merely shift Hamiltonian parameters in H_V without inducing information backflow. By condition (C1), the hidden sector is not decoupled: the interaction Hamiltonian H_{int} is continuously active. During each visible-sector transition, H_{int} imprints the identity of the transition onto the hidden-sector state. Because the hidden sector is slow — not static — these imprints persist without being thermally overwritten: the hidden sector's internal relaxation timescale (τ_B) vastly exceeds the interval between successive visible-sector transitions (τ_S). On subsequent transitions, the coupling reads back the stored correlations, producing transition probabilities that depend on the system's prior trajectory — the non-Markovian open-systems regime of Breuer and Petruccione [4].

2.3 P-Indivisibility

Theorem (P-indivisibility of embedded observation). *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define the one-step transition*

matrix on $\mathcal{C}_V$ by marginalizing over $\mathcal{C}_H$ with respect to the uniform measure:

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|}$$

and the k-step transition matrix $T_{ij}^{(k)}$ by the same construction applied to φ^k. If T is not a permutation matrix — i.e., the marginalized dynamics is genuinely stochastic — then the stochastic process generated by iterating φ is P-indivisible.

Proof. The proof uses the total variation distance $d(p, q) = \frac{1}{2}\sum_k |p_k - q_k|$ between probability distributions. A stochastic process is P-divisible if and only if $d(p(t), q(t))$ is non-increasing in t for all initial distributions p, q [6, 7]. P-indivisibility is established by exhibiting a non-monotonicity.

Step 1 (Recurrence). Since φ is a bijection on a finite set, it has finite order: there exists $N \geq 1$ such that $\varphi^N = \mathrm{id}$. Every state (x_i, h) returns to itself after N steps, so $T^{(N)} = I$. In particular, $d(\delta_i T^{(N)}, \delta_j T^{(N)}) = d(\delta_i, \delta_j) = 1$ for all $i \neq j$.

Step 2 (Strict contraction at step 1). Since T is row-stochastic and not a permutation matrix, there exist distinct indices i, j whose rows have overlapping support: some column l satisfies $T_{il} > 0$ and $T_{jl} > 0$. (If no two rows had overlapping support, their supports would be pairwise disjoint, forcing each row to concentrate on a single column — a permutation matrix, contradicting the hypothesis.) For such i, j:

$$d(\delta_i T, \delta_j T) = \frac{1}{2}\sum_k |T_{ik} - T_{jk}| < \frac{1}{2}\sum_k (T_{ik} + T_{jk}) = 1$$

The strict inequality holds because every column l with overlapping support contributes $|T_{il} - T_{jl}| < T_{il} + T_{jl}$ to the sum.

Step 3 (Non-monotonicity). Steps 1 and 2 together give:

$$d(\delta_i T^{(1)}, \delta_j T^{(1)}) < 1 = d(\delta_i T^{(N)}, \delta_j T^{(N)})$$

The total variation distance decreases from step 0 to step 1, then increases from step 1 to step N. This is non-monotonic: the process is P-indivisible. □

The theorem requires only two inputs: deterministic bijective dynamics (Axiom 1) and non-trivial coupling (condition C1, which ensures T is not a permutation matrix). Axiom 2 guarantees the finite configuration space on which the theorem operates. The proof mechanism is information backflow: information about the initial visible-sector state leaks to the hidden sector via the coupling (Step 2) and is later read back by the deterministic dynamics (Step 1), producing the non-monotonic trace distance that characterizes P-indivisibility.

Continuous-time extension. The theorem is stated for discrete-time dynamics, but the physical setup involves continuous-time Hamiltonian flow. The extension follows from continuity and Poincaré recurrence. The Hamiltonian flow φ_t on the finite-dimensional phase space (Axiom 2) preserves the Liouville measure on compact energy surfaces. The transition matrix $T_{ij}(t)$ is continuous in t, with $T(0) = I$. Non-zero coupling (C1) ensures that $T(t)$ departs from the permutation class for $t > 0$, so that $d(\delta_i T(t), \delta_j T(t)) < 1$ for

some pair i, j and some $t > 0$. By Poincaré's recurrence theorem, for any $\delta > 0$, there exists $t_R > 0$ such that the set of hidden-sector states satisfying $\pi_V(\varphi_{t_R}(x_i, h)) = x_i$ has measure $> 1 - \delta$, giving $d(\delta_i T(t_R), \delta_j T(t_R)) > 1 - 2\delta$. For δ small enough, this exceeds $d(\delta_i T(t), \delta_j T(t))$, establishing non-monotonic trace distance and hence P-indivisibility for the continuous-time process.

Convergence with prior work. The theorem converges with several independent lines of work: Pechukas [5] showed that reduced dynamics of a correlated system need not preserve positivity; Rivas, Huelga, and Plenio [6, 7] formalized the connection between divisibility failure and information backflow; Pollock et al. [8] derived necessary and sufficient Markovianity conditions via the process tensor framework; and most directly, Strasberg and Esposito [9] demonstrated non-Markovian reduced dynamics precisely in the slow-bath regime of condition (C2).

The physical role of conditions (C2) and (C3). The theorem establishes P-indivisibility from (C1) and finiteness alone. Conditions (C2) and (C3) are not needed for the mathematical result but are essential for its physical force. (C2) ensures that information backflow occurs on timescales accessible to the observer, not merely at Poincaré recurrence times that may exceed the age of the universe. (C3) ensures that P-indivisibility persists at physical scale: if the hidden sector has insufficient capacity, successive imprints overwrite earlier ones — the hidden sector saturates, and the reduced dynamics cross over to effectively Markovian behavior. The conjunction of (C2) and (C3) guarantees that P-indivisibility is strong, persistent, and observationally relevant.

2.4 Explicit Construction: A Minimal Model

The following minimal model illustrates the theorem and makes its mechanism explicit.

The model. The visible sector has two states: $x \in \{0, 1\}$. The hidden sector has six states: $h \in \{1, 2, 3, 4, 5, 6\}$. The total phase space has 12 states (x, h). The dynamics are a deterministic, invertible permutation σ on these 12 states, defined as six disjoint transpositions:

$$\sigma = (0{,}1 \leftrightarrow 1{,}1)\ (0{,}2 \leftrightarrow 1{,}2)\ (0{,}3 \leftrightarrow 0{,}4)\ (0{,}5 \leftrightarrow 0{,}6)\ (1{,}3 \leftrightarrow 1{,}4)\ (1{,}5 \leftrightarrow 1{,}6)$$

The first two transpositions couple the visible and hidden sectors — they change x while preserving h, implementing condition (C1). The remaining four permute hidden-sector labels within each visible-sector value, representing the hidden sector's internal dynamics. Since σ is a product of disjoint transpositions, $\sigma^2 = \mathrm{id}$: the hidden sector retains perfect correlations between successive transitions, satisfying condition (C2). Six hidden-sector states per visible-sector value ensures the hidden sector is not saturated by a single transition, satisfying condition (C3).

Transition matrices. At $t = 0$, the observer knows x but not h, and assigns the uniform (Liouville) measure over the 6 hidden states. At $t = 1$ (one application of σ):

From $x = 0$: two of six hidden states ($h = 1, 2$) send x to 1; four ($h = 3, 4, 5, 6$) leave x at 0. From $x = 1$: two of six ($h = 1, 2$) send x to 0; four leave x at 1. The one-step transition matrix is:

$$T(1,0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$$

At $t = 2$ ($\sigma^2 = \mathrm{id}$): every state returns to itself, so $T(2,0) = I$.

A Markov chain with transition matrix $T(1,0)$ would predict $T(2,0)_{\text{Markov}} = T(1,0)^2 = \left(\begin{smallmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{smallmatrix}\right)$ — continued mixing. The actual dynamics show complete *un-mixing*: the system returns to its initial state with certainty. This is information backflow.

P-divisibility test. If the process were P-divisible, one could write $T(2,0) = \Lambda(2,1) \cdot T(1,0)$ with $\Lambda(2,1)$ a valid stochastic matrix (non-negative entries, rows summing to 1). Computing:

$$\Lambda(2,1) = T(2,0) \cdot [T(1,0)]^{-1} = I \cdot \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

The matrix $\Lambda(2,1)$ has negative entries (-1). No valid stochastic matrix can serve as the intermediate propagator. **The process is P-indivisible.** □

The mechanism. The hidden sector acts as a memory register. At step 1, the transpositions $(0,h) \leftrightarrow (1,h)$ for $h \in \{1,2\}$ flip x while preserving h. After step 1, a state now at $x = 1$ with $h \in \{1,2\}$ carries the record "I was at $x = 0$." At step 2, the same transposition reads this record and flips x back. The visible-sector dynamics exhibit information backflow: the system appears to "remember" its origin, which is impossible for a memoryless process. The model illustrates the mechanism in the idealized limit of perfect recurrence ($\sigma^2 = \mathrm{id}$, i.e., $\tau_B/\tau_S = 2$). The cosmological realization of Part II has $\tau_S/\tau_B \sim 10^{-32}$; the P-indivisibility is more extreme, not less, when the hidden sector's memory is more persistent.

3. THE EMERGENCE OF QUANTUM MECHANICS

3.1 The Stochastic-Quantum Correspondence

By Barandes' stochastic-quantum correspondence [10, 11], any indivisible stochastic process on a finite configuration space of size n can be embedded into a unitarily evolving quantum system on a Hilbert space of dimension $\leq n^3$. A stochastic process on n configurations is characterized by its transition matrices $T_{ij}(t_2, t_1)$. If the process is P-divisible, these matrices can be factored through intermediate times and the system admits a classical description at all scales. If it is P-indivisible, no such factoring exists: the transition matrices at fine time resolution develop negative entries (as in §2.4), violating classicality. Barandes shows that the Stinespring dilation theorem then guarantees the existence of a Hilbert space and a unitary operator $U(t)$ such that $T_{ij}(t) = |U_{ij}(t)|^2$ — the transition probabilities of the stochastic process are exactly the Born-rule probabilities of the quantum system. Indivisibility is not merely *analogous* to quantum behavior; it is *mathematically equivalent* to it. The P-indivisible process established in §2.3 satisfies the correspondence's hypotheses directly.

From phase space to configuration space. The axioms define the dynamics on a phase space Γ with coordinates (q, p), while Barandes' correspondence operates on a

configuration space. The connection is the marginalization itself: the transition probabilities $T_{ij}(t_2, t_1)$ defined in §2.1 are projections of the full phase-space flow onto the configuration sector of Γ_V, with the momenta — along with all hidden-sector degrees of freedom — absorbed into the marginalization over the Liouville measure. The resulting stochastic process lives on the discrete configuration space defined by Axiom 2, where the correspondence applies directly.

Three features of quantum mechanics emerge from this correspondence. The Schrödinger equation is the unique time-evolution law, arising from the differentiability of the unitary in time. The Born rule is not an additional postulate but the equilibrium distribution of the indivisible stochastic process [10, 11]. The action scale $\hbar$ requires more careful treatment. The correspondence guarantees the existence of a unitary evolution operator $U(t)$, but U is dimensionless; $\hbar$ enters only when one defines the Hamiltonian $\hat{H}(t) \equiv i\hbar\,\partial_t U \cdot U^\dagger$, converting the dimensionless rate of change into energy units. The correspondence itself is silent about the value of this conversion factor — it establishes that a quantum description exists for any P-indivisible process, but not which action scale it carries. What fixes $\hbar$ is the physical content of the partition: the emergent description must be unique for a given partition (§1.4), so the action scale can depend only on quantities intrinsic to the partition boundary. Its value is derived for the cosmological realization in Part II (§5).

Finite-dimensionality. Barandes' proof requires a finite-dimensional configuration space. In Part I, Axiom 2 guarantees this directly. In Part II, the horizon's finite area bounds the coupled modes to A/ϵ^2. Calvo [12] has extended the correspondence to infinite-dimensional systems, but the extension is not required here.

3.2 Bell Inequality Violations

The framework is a hidden variable theory: quantum randomness arises epistemically from ignorance of hidden-sector degrees of freedom. Bell's theorem [13] rules out *local* hidden variable theories satisfying the factorizability condition $P(a, b \mid x, y, \lambda) = P(a \mid x, \lambda) \cdot P(b \mid y, \lambda)$. The present framework evades Bell's theorem not by violating measurement independence (superdeterminism) but by violating factorizability through P-indivisible joint dynamics — while remaining causally local.

The mechanism is non-factorizable joint dynamics from a shared causal past. Two subsystems Q and R that interact at preparation and are subsequently separated carry a joint transition matrix $T_{QR}(t_2, t_1)$ on $\mathcal{C}_Q \times \mathcal{C}_R$ that does not factorize:

$$T_{QR} \neq T_Q \otimes T_R$$

This non-factorizability *is* entanglement at the stochastic level [10, 11, 14]. The correlations are encoded in the structure of the joint transition matrix rather than in a hidden variable λ over which one could integrate. Since the process is indivisible, there are no well-defined intermediate conditional probabilities that would permit the factorization Bell's theorem requires.

In the Jarrett decomposition, quantum mechanics violates outcome independence (OI) while preserving parameter independence (PI). The present framework reproduces this structure exactly. Barandes, Hasan, and Kagan [14] prove that the conjunction of indivisibility and causal locality restricts the maximum CHSH correlator to exactly Tsirelson's

bound $|S| \leq 2\sqrt{2}$, with independent support from Le et al. [15]. Measurement independence is preserved by construction: transition probabilities are defined independently of the initial distribution. This placement — violating outcome independence while preserving measurement independence and parameter independence — is precisely the class Fine's theorem [24] permits.

3.3 The Characterization Theorem: Necessity of the Conditions

The theorem of §1.3 establishes sufficiency: Axioms 1–4 and conditions (C1)–(C3) guarantee quantum mechanics. This section proves the converse — that quantum mechanics *requires* these conditions — establishing a full equivalence between quantum mechanics and embedded observation.

The logical structure. Barandes' stochastic-quantum correspondence is already bidirectional: a stochastic process is P-indivisible if and only if it is mathematically equivalent to unitarily evolving quantum mechanics [10, 11]. The equivalence

$$\text{QM} \iff \text{P-indivisibility}$$

is established. What remains is to prove the converse of the other link:

$$\text{P-indivisibility} \implies \text{embedded observation under (C1)–(C3)}$$

The full equivalence QM $\iff$ embedded observation (under Axioms 1–4 and (C1)–(C3)) then follows by transitivity.

Dilation existence (necessity of the axioms). Any stochastic process on a finite configuration space can be realized as the marginalization of a deterministic process on a larger state space: one enlarges the state space to encode the randomness as ignorance of auxiliary variables and recovers the original process by tracing them out. This is a standard construction in classical probability theory. The dilation furnishes a deterministic total system (Axiom 1), a finite configuration space (Axiom 2), a partition into visible and hidden sectors (Axiom 3), and classical marginalization (Axiom 4). The four axioms are therefore necessary in the sense that any stochastic process — and hence any P-indivisible process — admits a deterministic embedding with a causal partition on a finite configuration space.

Theorem (C1 equivalence). *The marginalized process is P-indivisible if and only if T is not a permutation matrix (i.e., C1 holds).*

Proof. Forward: §2.3. Reverse: if T is a permutation matrix, then $T^{(k)}$ is also a permutation for every k, and $\Lambda(k_2, k_1) = T^{(k_2-k_1)}$ is a valid stochastic matrix for all $k_2 > k_1$. The process is P-divisible. Contrapositively, P-indivisibility requires non-trivial coupling. □

Theorem (C2 necessity: Markovian limit). *Let φ_{coup} implement the visible-hidden coupling and φ_H be an ergodic permutation on $\mathcal{C}_H$. In the fast-bath regime ($\tau_B \ll \tau_S$), ergodic mixing drives the hidden-sector distribution to uniformity before each coupling event, making the hidden sector's state independent of the visible sector's history. The resulting process is a Markov chain with $T^{(k)} = T^k$, and the intermediate propagator $\Lambda(k_2, k_1) = T^{k_2-k_1}$ is a valid stochastic matrix for all $k_2 > k_1$. The process is P-divisible.*

Proof sketch. If the hidden sector re-randomizes between coupling events, the k-step matrix is the k-th power of the one-step matrix — a Markov chain, which is P-divisible. Fast mixing ($m \gg |\mathcal{C}_H|$ hidden-sector steps between couplings) drives the conditioned hidden-sector distribution to uniformity by the ergodic theorem for finite groups. Contrapositively, P-indivisibility requires the hidden sector to retain memory of past visible-sector transitions, which requires $\tau_S \ll \tau_B$ — condition (C2). □

A slow bath retains imprints of past transitions and reads them back on subsequent steps (information backflow). A fast bath erases those imprints through internal mixing, destroying the memory that P-indivisibility requires.

Theorem (C3 necessity: information-theoretic capacity bound). *Let φ be a bijection on $\mathcal{C}_V \times \mathcal{C}_H$ with $|\mathcal{C}_V| = n$ and $|\mathcal{C}_H| = m$. Let $X_0, X_1, X_2, \ldots$ denote the visible-sector trajectory and $H_0, H_1, H_2, \ldots$ the corresponding hidden-sector trajectory under the deterministic dynamics. The non-Markovian mutual information — the information that the visible sector's history provides about its future evolution, beyond what is contained in its present state — satisfies:*

$$I(X_{<t}; X_{>t} \mid X_t) \leq \log_2 m$$

In particular, the total non-Markovian information capacity of the reduced process is bounded by the logarithmic size of the hidden sector.

Proof. The total system evolves deterministically: X_{t+1} and H_{t+1} are deterministic functions of (X_t, H_t). Therefore the entire future trajectory $X_{>t} = (X_{t+1}, X_{t+2}, \ldots)$ is a deterministic function of (X_t, H_t). By the data processing inequality for Markov chains: if $A \to B \to C$ is a Markov chain, then $I(A; C) \leq I(A; B)$. Conditioning on X_t:

$$X_{<t} \to H_t \to X_{>t} \quad \text{(conditioned on } X_t\text{)}$$

This is a Markov chain because, given X_t and H_t, the future $X_{>t}$ is fully determined and independent of $X_{<t}$. Applying the data processing inequality:

$$I(X_{<t}; X_{>t} \mid X_t) \leq I(X_{<t}; H_t \mid X_t) \leq H(H_t \mid X_t) \leq \log_2 |\mathcal{C}_H| = \log_2 m$$

where the second inequality is a standard property of mutual information ($I(A; B) \leq H(B)$), and the third is the maximum entropy of a random variable on m values. □

Corollary (necessity of C3). *P-indivisibility across all n visible-sector configurations requires at least $\log_2 n$ bits of hidden-sector memory. By the capacity bound, $m \geq n$, and persistence over K successive transitions requires m/n to grow with the observation window. The hidden sector must scale with the visible sector — precisely condition (C3).* □

Characterization theorem (equivalence of quantum mechanics and embedded observation). *Let $\mathcal{C}_V$ be a finite configuration space with $|\mathcal{C}_V| \geq 2$, and let $\mathcal{S}$ be a stochastic process on $\mathcal{C}_V$. The following are equivalent:*

(i) $\mathcal{S}$ is mathematically equivalent to unitarily evolving quantum mechanics (via Barandes' stochastic-quantum correspondence [10, 11]).

(ii) $\mathcal{S}$ is P-indivisible.

(iii) $\mathcal{S}$ arises from the marginalization of a deterministic bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ over a hidden sector $\mathcal{C}_H$, with: - (C1) non-trivial coupling: the marginalized transition matrix T is not a permutation matrix; - (C2) slow-bath memory: the hidden sector retains correlations between successive coupling events (its mixing timescale exceeds the visible sector's dynamical timescale); - (C3) sufficient capacity: $|\mathcal{C}_H| \geq |\mathcal{C}_V|$, with $|\mathcal{C}_H|/|\mathcal{C}_V|$ large enough that the hidden sector is not saturated by visible-sector interactions over the relevant observation window.

Proof. (i) $\iff$ (ii) is Barandes' stochastic-quantum correspondence [10, 11]. (iii) $\implies$ (ii) is the theorem of §2.3. (ii) $\implies$ (iii) is proved by the three necessity theorems above. □

Interpretive consequence. The characterization theorem closes the logical gap that a sufficiency-only result would leave open. Quantum mechanics is not merely *compatible with* the epistemic situation of an embedded observer — it is *equivalent to* it. Every quantum system can be understood as one in which an observer is embedded in a larger deterministic system with non-trivial, slowly decorrelating coupling to a hidden sector of sufficient capacity. Conversely, every such embedding necessarily produces quantum mechanics in the observer's reduced description.

PART II: THE COSMOLOGICAL APPLICATION

4. THE COSMOLOGICAL HORIZON AS CAUSAL PARTITION

4.1 The Partition

General relativity defines the causal structure of spacetime. The cosmological horizon is the boundary beyond which no signal propagating at or below c can reach the embedded observer. Every physically realizable measurement technique — electromagnetic, gravitational-wave, neutrino — propagates at or below c and therefore encounters the same boundary. The horizon implements the causal partition of Axiom 3 for all sub-c observers simultaneously: the visible sector Γ_V is the interior of the cosmological horizon, and the hidden sector Γ_H is everything beyond it.

This identification is not a modeling choice but a consequence of GR's causal structure. A hypothetical observer with access to a causal channel not constrained by the light cone would have a different partition, a different (or absent) hidden sector, and by the theorem's own logic, a different emergent description. No such channel is available within GR. The specific quantum mechanics observed is the quantum mechanics of light-cone-bounded observers; its universality is a consequence of the causal structure, not a postulate.

4.2 Verification of the Conditions

Condition (C1): Non-zero coupling. The cosmological horizon is not a static wall. Stress-energy conservation across the horizon enforces dynamical correlations between the visible and hidden sectors: matter and radiation cross the horizon continuously, and

the horizon area responds to changes in the interior energy density via the Friedmann equation. The coupling is bidirectional. **Condition (C1) is satisfied.**

Condition (C2): Slow-bath timescale separation. The hidden sector's shortest correlation timescale is set by the Hubble time: $\tau_B \gtrsim 1/H \sim 10^{17}$ s, since modes beyond the horizon evolve on timescales no shorter than the rate at which the horizon itself changes. (Modes deeper in the hidden sector have even longer correlation times.) For any laboratory-scale process with characteristic frequency $\omega \gg H$, the visible sector's dynamical timescale is $\tau_S \sim 1/\omega \sim 10^{-15}$ s or shorter. The ratio is:

$$\frac{\tau_S}{\tau_B} \sim 10^{-32}$$

The hidden sector cannot decorrelate between any pair of visible-sector events. The Markovian approximation fails categorically. **Condition (C2) is satisfied.**

Condition (C3): Sufficient capacity. The hidden sector has $N_{\text{modes}} = A/\epsilon^2$ independent degrees of freedom, where $A = 4\pi c^2/H^2$ is the horizon area and ϵ is the discreteness scale of Axiom 2. For any ϵ at or above the Planck scale, the horizon area is cosmological ($A \sim 10^{52}$ m^2) and the visible sector involves at most $\sim 10^{80}$ baryons. No laboratory process can appreciably perturb the hidden sector's state. The hidden sector is never saturated. **Condition (C3) is satisfied.**

4.3 Application of the Theorem

All four axioms hold. All three conditions hold. Axiom 2 — which is an independent assumption in Part I — is here a consequence of the partition geometry: the cosmological horizon has finite area $A = 4\pi c^2/H^2$, and all visible-hidden coupling passes through this boundary. The number of modes that participate in the trace-out is bounded by A/ϵ^2, which is finite for any $\epsilon > 0$. The effective stochastic process therefore lives on a finite-dimensional configuration space, as Barandes' correspondence requires, without any assumption that the total phase space beyond the horizon is itself finite.

The theorem of Part I applies: the embedded observer's reduced description of the visible sector is P-indivisible, and by Barandes' stochastic-quantum correspondence [10, 11], mathematically equivalent to unitarily evolving quantum mechanics with an action scale $\hbar$ determined by the local structural properties of the partition boundary.

What this step requires: Part I plus Barandes. Axiom 2 is satisfied by the partition geometry. No additional physics.

5. DERIVATION OF THE EMERGENT ACTION SCALE ($\hbar$)

5.1 The Classical Horizon Temperature

The theorem guarantees that $\hbar$ exists but does not compute its value. Partition-relativity (§1.4) constrains $\hbar$ to depend only on quantities intrinsic to the partition — the geometric and thermodynamic properties of the boundary between visible and hidden sectors. The required thermodynamic input is model-independent: it follows from Jacobson's thermodynamic derivation of Einstein's field equations [16].

Jacobson (1995) demonstrated that applying the thermodynamic relation $dE = T\,dS$ to local causal horizons yields Einstein's field equations. The same identity relates mass-energy to area via:

$$dE = \frac{c^2\kappa}{8\pi G}\,dA$$

The classical entropy density on the horizon is $\eta = 1/\epsilon^2$ — one degree of freedom per minimal cell (Axiom 2). Therefore $dS = dA/\epsilon^2$, and equating $dE = T_{\rm cl}\,dS$ fixes the classical temperature:

$$T_{\rm cl} = \frac{c^2\epsilon^2\kappa}{8\pi G k_B}$$

For the de Sitter horizon, the surface gravity is $\kappa = cH$, giving:

$$k_B T_{\rm cl} = \frac{c^3\epsilon^2 H}{8\pi G}$$

This temperature depends on G, c, H, ϵ, and k_B — no $\hbar$. It is the temperature of the hidden sector as a classical thermal reservoir, derived from the horizon geometry without reference to quantum mechanics.

5.2 The Emergent Action Scale

The stochastic-quantum correspondence maps the P-indivisible transition matrices to a unitary evolution $U(t)$, but leaves an enormous gauge freedom: the transition matrix constrains only $|U_{ij}|^2$, leaving the quantum phases — and hence the energy eigenvalues and the overall action scale — undetermined. This is the Schur-Hadamard gauge freedom inherent in the stochastic-to-quantum map. Partition-relativity (§1.4) constrains the resolution: the emergent quantum description is uniquely determined by the partition, so the action scale $\hbar$ can depend only on quantities intrinsic to the partition — the geometric and thermodynamic properties of the boundary between visible and hidden sectors.

The determination of $\hbar$ proceeds in four steps. Steps 1–3 establish the scaling; Step 4 fixes the prefactor. Step 2 invokes a result proved in §5.3; the derivation is self-contained once that section is reached.

Step 1: Uniqueness from partition-relativity. By the lemma of §1.4, the three inputs — dynamics ϕ_t, Liouville measure μ, and partition $\Gamma_V \times \Gamma_H$ — yield a single stochastic process, which maps to a single quantum description. The action scale $\hbar$ is therefore a derived quantity, not a free parameter.

Step 2: The action scale is structural, not volumetric. The action scale governs the stochastic-to-unitary conversion — how the hidden sector couples to and imprints correlations on the visible sector — not the total capacity of the hidden sector. The distinction is analogous to a classical thermal reservoir: temperature determines the Boltzmann factor regardless of the bath's total size. If $\hbar$ depended on H through volumetric factors, observers at different cosmological epochs would obtain different quantum mechanics.

This argument is made precise by the following factorization result, combined with the D-gauge completeness theorem of §5.3.

Lemma (volume-independence of transition probabilities). *Let $(\mathcal{C}_V, \mathcal{C}_H, \varphi_t)$ be a deterministic system with transition matrix $T_{ij}(t)$ on $\mathcal{C}_V$. Decompose $\mathcal{C}_H = \mathcal{C}_B \times \mathcal{C}_D$ into boundary modes $\mathcal{C}_B$ and deep modes $\mathcal{C}_D$. If the visible-sector transition is independent of the deep modes on timescales up to t_* — that is, $\pi_V(\varphi_t(x, h_B, h_D)) = \pi_V(\varphi_t(x, h_B, h'_D))$ for all h_D, h'_D and all $t \leq t_*$ — then $T_{ij}(t) = T^{(B)}_{ij}(t)$ for all $t \leq t_*$, where $T^{(B)}$ is computed by marginalizing over $\mathcal{C}_B$ alone.*

Proof. For $t \leq t_*$:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B||\mathcal{C}_D|} \sum_{h_B, h_D} \mathbf{1}[\pi_V(\varphi_t(x_i, h_B, h_D)) = x_j]$$

By hypothesis, the indicator function is independent of h_D, so the sum over $\mathcal{C}_D$ contributes a factor of $|\mathcal{C}_D|$ that cancels the $1/|\mathcal{C}_D|$ prefactor:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B|} \sum_{h_B} \mathbf{1}[\pi_V(\varphi_t(x_i, h_B)) = x_j] = T^{(B)}_{ij}(t) \quad \square$$

The lemma extends to approximate independence: if the visible-sector transition depends on the deep modes for at most a fraction δ of configurations, then $|T_{ij}(t) - T^{(B)}_{ij}(t)| \leq \delta$.

Application to the cosmological partition. The boundary modes are the A/ϵ^2 degrees of freedom on the horizon surface; the deep modes are degrees of freedom at distances $\gg c/H$ beyond the horizon. Deep modes can influence the boundary only by propagating through the hidden sector, which takes time $\gtrsim \tau_B$ (C2). For all $t \ll \tau_B$, the lemma's condition is satisfied with $\delta \lesssim t/\tau_B \sim 10^{-32}$, and the transition probabilities are volume-independent.

Step 3: Dimensional determination. With volumetric quantities excluded, the partition-intrinsic quantities that can appear in $\hbar$ are restricted to those characterizing the local structure of the boundary: c (defines the causal boundary), G (enters through Jacobson's identity relating geometry to energy), and ϵ (sets the phase-space grain). The unique combination with dimensions of action is $c^3\epsilon^2/G$, giving:

$$\hbar = \beta \frac{c^3 \epsilon^2}{G}$$

where β is a dimensionless prefactor.

Step 4: Thermal self-consistency fixes the prefactor.

Before the trace-out (§5.1), the classical horizon temperature is derived from Jacobson's thermodynamic identity and the entropy density $\eta = 1/\epsilon^2$:

$$k_B T_{\text{cl}} = \frac{c^3 \epsilon^2 \kappa}{8\pi G}$$

This expression involves only classical quantities: G, c, ϵ, κ, k_B. No $\hbar$ appears.

After the trace-out, the emergent quantum mechanics is established by Part I with an action scale $\hbar$ whose scaling is $\hbar = \beta c^3\epsilon^2/G$ (Step 3) but whose prefactor β is not yet fixed. Quantum field theory on curved spacetime — which is now a *consequence* of the emergent description, not an independent input — predicts the Gibbons-Hawking temperature [17] for any horizon with surface gravity κ:

$$k_B T_{\mathrm{GH}} = \frac{\hbar\kappa}{2\pi c}$$

These two temperatures describe the same physical boundary. T_{cl} is the temperature of the hidden sector as a classical thermal reservoir, derived from the partition geometry. T_{GH} is the temperature that the emergent quantum description assigns to the same horizon. If $T_{\mathrm{cl}} \neq T_{\mathrm{GH}}$, the emergent description is internally inconsistent: it predicts a different thermal state for the very boundary that produced it.

Self-consistency requires $T_{\mathrm{cl}} = T_{\mathrm{GH}}$. For the de Sitter horizon ($\kappa = cH$):

$$\frac{c^3\epsilon^2 H}{8\pi G} = \frac{\hbar H}{2\pi}$$

The Hubble parameter H cancels — confirming the structural/volumetric point, since the result is H-independent. Solving for $\hbar$:

$$\boxed{\hbar = \frac{c^3\epsilon^2}{4G}}$$

The prefactor is $\beta = 1/4$. No free parameter remains.

Why this is not circular. The argument uses the Gibbons-Hawking temperature, which is a result of quantum field theory on curved spacetime — the very theory whose emergence the theorem establishes. The logic is sequential, not circular:

1. Part I establishes that the emergent description *is* quantum mechanics, with *some* action scale $\hbar$ (existence, not value).
2. The emergent QM, applied to the de Sitter horizon, predicts $T_{\mathrm{GH}}(\hbar) = \hbar H/(2\pi k_B)$. This is a *consequence* of the emergent theory, expressed in terms of the unknown $\hbar$.
3. The classical temperature T_{cl} is derived independently in §5.1, with no $\hbar$.
4. Matching the two — requiring that the emergent description be consistent with the conditions that produced it — gives one equation in one unknown.

This is a *gap equation*: the same logical structure as self-consistent mean-field calculations in condensed matter physics, where the order parameter is determined by requiring the emergent collective description to reproduce the microscopic conditions that generated it. The Gibbons-Hawking temperature is not imported as an external postulate; it is a *prediction* of the emergent theory that must be consistent with the classical temperature that preceded the theory's emergence.

Robustness. The self-consistency condition can be derived without the Gibbons-Hawking formula: the hidden sector is a thermal reservoir at T_{cl}, the visible sector equilibrates with it (C1–C3 ensure persistent thermal contact), and the zeroth law of

quantum statistical mechanics gives $T_Q = T_{\rm cl}$ exactly. The Gibbons-Hawking formula is then *recovered* as a prediction. Lattice corrections enter only as $\mathcal{O}((\epsilon H/c)^2) \sim 10^{-122}$, consistent with trans-Planckian insensitivity [28, 29].

5.3 Gauge-Fixing, the Dimensional Obstruction, and the Role of Thermal Self-Consistency

The stochastic-quantum correspondence maps the P-indivisible process to a unitary $U(t)$, but leaves the Schur-Hadamard gauge freedom: the transition probabilities $T_{ij} = |U_{ij}|^2$ do not uniquely determine the quantum phases. Continuous-time transition data resolves this ambiguity. The following theorem shows that matching the full time-evolution forces the phases into a unique configuration, up to a physically trivial rephasing.

Theorem (D-gauge completeness). *Let $U(t) = e^{-iHt}$ be a unitary propagator with non-degenerate eigenvalues and non-vanishing configuration-basis overlaps. If $U'(t) = e^{-iH't}$ satisfies $|U'_{ij}(t)|^2 = |U_{ij}(t)|^2$ for all i, j and all t, then $H' = DHD^\dagger$ for a time-independent diagonal unitary $D = diag(e^{i\alpha_1}, \ldots, e^{i\alpha_n})$.* (Proof: expanding $|U_{ij}(t)|^2$ as a Fourier series in the energy differences, non-degeneracy uniquely determines the moduli $|c_{ia}|$; matching the phase constraints forces $\delta_{ia} = \alpha_i + \beta_a$, giving $H' = DHD^\dagger$.)

The D-gauge is physically trivial: it corresponds to a time-independent basis rephasing that leaves all transition probabilities, energy-level spacings, and interference patterns invariant. The non-degeneracy conditions hold generically (measure zero among Hermitian matrices).

Cross-interval matching and time dependence. For time-dependent effective generators (condition C2 ensures $H_{\rm eff}(t)$ is approximately time-independent on intervals $\tau_S \ll \Delta t \ll \tau_B$), the D-gauge theorem applies within each interval. Cross-interval transition probabilities fix the relative gauge between intervals up to a single global phase: the requirement $|\sum_j e^{i\phi_j} U_{ij} U_{jk}|^2 = |\sum_j U_{ij} U_{jk}|^2$ for all i, k forces $\phi_j = \phi$ for all j. In the continuum limit, the chain of global phases becomes a smooth function $e^{i\theta(t)}$ with no observable consequences. The continuous-time transition probabilities therefore determine the unitary propagator up to physically irrelevant rephasing.

The dimensional obstruction. The preceding results resolve all *dimensionless* ambiguity but leave $\hbar$ untouched — by structural necessity. The unitary $U(t)$ is dimensionless; $\hbar$ enters only when defining $\hat{H} = i\hbar\,\partial_t U \cdot U^\dagger$, converting dimensionless rates into energy units. No quantity of dimensionless probability data can fix a dimensionful constant. Step 4 provides the necessary dimensionful input: the thermal self-consistency condition $T_{\rm cl} = T_{\rm GH}(\hbar)$ connects the dimensionless oscillation frequencies to a physical temperature derived from the classical partition geometry (§5.1).

6. SELF-CONSISTENCY AND THE DISCRETENESS SCALE

Rearranging $\hbar = c^3\epsilon^2/(4G)$:

$$\epsilon^2 = \frac{4\hbar G}{c^3} = 4\,l_p^2$$

The framework predicts $\epsilon = 2\,l_p$ exactly. The geometric discreteness scale is twice the Planck length — a specific quantitative prediction, not an order-of-magnitude estimate.

This value is necessary for self-consistency. If $\epsilon^2 \ll l_p^2$, sub-Planckian subcells would be dynamically active yet unresolvable by the emergent description, creating a second trace-out with its own noise intensity and breaking the single-valuedness of $\hbar$. If $\epsilon^2 \gg l_p^2$, the emergent description would assign distinct quantum states to configurations that are physically identical in the substratum. The regime $\epsilon = \mathcal{O}(l_p)$ is the only self-consistent one; thermal self-consistency selects $\epsilon = 2\,l_p$ within it.

With $\epsilon = 2\,l_p$, the number of independent modes on the cosmological horizon is:

$$N_{\text{modes}} = \frac{A}{\epsilon^2} = \frac{A}{4\,l_p^2} = S_{\text{dS}}$$

where $S_{\text{dS}} = A/(4\,l_p^2)$ is the Bekenstein-Hawking entropy of the de Sitter horizon. The factor of 1/4 in the Bekenstein-Hawking formula — historically introduced as a proportionality constant — is here derived as a consequence of the partition geometry and the thermal self-consistency condition: each minimal cell of area $\epsilon^2 = 4\,l_p^2$ on the horizon contributes one unit of entropy, and $A/(4\,l_p^2) = A/\epsilon^2$ recovers the standard formula exactly.

Consistency check. After the emergence of quantum mechanics and the identification $\epsilon = 2\,l_p$, the classical horizon temperature becomes $T_{\text{cl}} = c^3\epsilon^2\kappa/(8\pi G k_B) = c^3(4\,l_p^2)\kappa/(8\pi G k_B) = \hbar\kappa/(2\pi c k_B)$ — the standard Gibbons-Hawking temperature [17] exactly. This recovery is guaranteed by construction (it is the self-consistency condition of Step 4), but verifying it confirms the algebraic consistency of the derivation.

7. DISSOLUTION OF THE COSMOLOGICAL CONSTANT PROBLEM

7.1 The Two Levels of Description

The cosmological constant problem [2, 3, 18] — the 10^{122}-fold discrepancy between the QFT prediction for vacuum energy density and the observed value — is often called the worst prediction in physics. Its resolution turns on recognizing that the classical substratum and the emergent quantum description give different — and independently self-consistent — answers to the same question.

From finite-dimensional QM to QFT. The theorem of Part I delivers quantum mechanics on a finite-dimensional Hilbert space. The connection to quantum field theory proceeds through three standard steps. First, Axiom 2 discretizes the visible sector into $N = A/\epsilon^2$ cells of area ϵ^2, each coupled to the hidden sector through the partition boundary. Second, the causal structure of the classical substratum restricts which cells interact on any given timescale, and the emergent quantum description inherits this locality. Third, spatial locality implies the emergent Hilbert space decomposes as $\mathcal{H} = \bigotimes_k \mathcal{H}_k$ — a lattice-regularized QFT with a built-in ultraviolet cutoff at $\epsilon = 2\,l_p$. The zero-point energy $\frac{1}{2}\hbar\omega$ per mode and the resulting QFT vacuum energy density are properties of this emergent description, not of the classical substratum. The third step — the tensor product decomposition — is standard for spatially local Hamiltonian systems on a lattice,

but a rigorous derivation from the axioms requires showing that the Barandes correspondence preserves spatial locality. This is expected to hold generically but is not proved here; the CC dissolution rests on this assumption.

The classical substratum (what geometric measurements probe): The cosmological horizon has a classical thermal equilibrium. By the Friedmann equation, $H^2 = (8\pi G/3)\rho$, the horizon area $A = 4\pi c^2/H^2$ adjusts self-consistently so that the mean energy density matches the critical density $\rho_{\text{crit}} = 3H^2c^2/(8\pi G)$. The classical energy per mode is $k_B T_H$. There is no zero-point energy and no discrepancy. The total vacuum energy density is at the critical scale $\rho \sim H^2/G \sim 10^{-9}$ J/m^3 — precisely what is observed.

The emergent QFT (what local quantum measurements probe): Quantum field theory assigns a zero-point energy of $\frac{1}{2}\hbar\omega$ per mode. Summing to the Planck cutoff yields $\rho_{\text{QFT}} \sim M_{\text{Pl}}^4 \sim 10^{113}$ J/m^3.

7.2 Why Gravity Sees the Classical Value

Spacetime geometry is part of the classical substratum (Axiom 3): the metric tensor is defined on the fundamental phase space and evolves via Einstein's field equations *before* the trace-out that produces quantum mechanics. The stress-energy tensor that sources the Einstein equations is the classical stress-energy of the total microstate, not the expectation value of an emergent quantum operator. The 10^{113} J/m^3 zero-point energy exists in the observer's informational ledger — it is a consequence of re-describing the classical noise as quantum fluctuations — but it does not appear in the stress-energy tensor that governs the geometry. The semiclassical gravity equation $G_{\mu\nu} = 8\pi G\langle\hat{T}_{\mu\nu}\rangle$ is itself an emergent approximation, valid when the quantum description is treated as fundamental. Within the present framework, where the quantum description is derived, the gravitational field equations operate at the classical level and never encounter the zero-point sum.

Ontological commitment and its consequences. The framework makes a specific ontological commitment: classical spacetime geometry is fundamental; quantum mechanics is emergent. This is not a free choice but follows from the logical structure of the derivation. The causal partition that produces quantum mechanics is defined by null geodesics of the metric (§4.1); the metric must therefore exist prior to the partition, and hence prior to the quantum description the partition generates. Reversing this ordering — treating the metric as emergent from quantum mechanics — would make the partition definition circular. This logical priority is opposite to programs that treat the metric as emergent (canonical quantum gravity, asymptotic safety, string-theoretic emergent spacetime), and the ordering question has observable consequences.

The cosmological constant problem is one such consequence. Any framework in which the quantum description is logically prior to the metric will necessarily treat the zero-point sum as a source term in the gravitational field equations, because the stress-energy tensor is constructed from quantum operators. The 10^{122} discrepancy is then a real problem requiring cancellation or fine-tuning. In the present framework, where the metric is logically prior, the zero-point sum is an artifact of the emergent description and never enters the stress-energy tensor. The dissolution of the CC problem is therefore a direct prediction of the framework's central commitment — not a separate result, and not an ad hoc stipulation, but a consequence of taking seriously the derivation's own logical structure.

Quantum corrections to gravity. This exclusion of the zero-point sum does not, however, mean that gravity is immune to all quantum effects. The emergent quantum description does feed back into the gravitational dynamics through state-level quantities: §8.1 derives dark energy evolution in Running Vacuum form precisely because the emergent vacuum energy — a state-level property of the quantum description — depends on the Hubble parameter through the hidden sector's volume. This H-dependent running of Λ_{eff} *is* a quantum correction to gravity.

What the framework excludes is the zero-point sum — a property of the correspondence itself rather than of any particular state — as a gravitational source. The structural/volumetric distinction of §5.2 applies here too: the action scale $\hbar$ (structural) does not gravitate; the vacuum energy (volumetric, state-dependent) does, but at the classical scale $\rho \sim H^2/G$ rather than the QFT scale $\rho \sim M_{\text{Pl}}^4$.

7.3 The Structural Origin of the 10^{122} Discrepancy

The ratio $\rho_{\text{QFT}}/\rho_{\text{classical}} \sim M_{\text{Pl}}^4/(H^2/G) = S_{\text{dS}}$. The discrepancy equals $S_{\text{dS}} = A/(4\,l_p^2)$ — the Bekenstein-Hawking entropy of the cosmological horizon, now identified as the number of independent modes on the partition boundary ($N_{\text{modes}} = A/\epsilon^2 = S_{\text{dS}}$). This is the information compression ratio of the quantum description: the number of hidden-sector degrees of freedom that the trace-out compresses into the emergent quantum state. The "worst prediction in physics" is the entropy of the observer's blind spot — a category error, not a fine-tuning failure.

That this discrepancy cannot be resolved from within is confirmed by Wolpert's (2008) limits of inference [19]: any physical subsystem whose state is a deterministic, many-to-one function of the total configuration is subject to absolute inferential limits. Both geometric measurements (classical substratum) and local particle measurements (emergent QFT) are faithful to their respective descriptions; no embedded observer can determine from within which is more fundamental.

This is not a prediction awaiting future data. The observed vacuum energy sits at the classical geometric scale $\rho \sim H^2/G$ — the value the framework expects. Five decades of searching for a cancellation mechanism within frameworks where quantum mechanics is fundamental have found nothing (supersymmetric cancellations increasingly constrained by LHC null results, anthropic arguments unable to predict the specific observed value). The framework explains why: there is nothing to cancel.

8. PREDICTIONS AND TESTS

8.1 Dark Energy Evolution in Running Vacuum Form

The action scale $\hbar$ depends only on structural properties of the partition boundary and is independent of H (§5.2). But the emergent vacuum energy is a state-level quantity that depends on the hidden sector's volume. The horizon area $A = 4\pi c^2/H^2$ evolves with the Hubble parameter, so the emergent vacuum energy inherits a dependence on H. Expanding $\Lambda_{\text{eff}}(H)$ around the present epoch:

$$\Lambda_{\text{eff}}(H) = \Lambda_0 + \frac{3\nu_{\text{OI}}}{8\pi G}(H^2 - H_0^2) + \mathcal{O}((H^2 - H_0^2)^2)$$

This is precisely the form of the Running Vacuum Model (RVM) [20, 21]. The framework requires $\nu_{\rm OI} > 0$ (the hidden sector gains capacity as H decreases) and $\nu_{\rm OI} \ll 1$ (condition C3 requires the hidden sector not be saturated). Its value is determined by the spectral structure of the partition boundary.

The $S_{\rm dS}$ horizon modes span frequencies from $\omega_{\rm IR} = H$ to $\omega_{\rm UV} = c/\epsilon$. Massless field modes on a 2-sphere have conformal mode density — equal numbers of independent modes per logarithmic frequency interval — a consequence of the conformal invariance of the wave equation on the 2-sphere, which maps modes at different scales into one another without introducing a preferred frequency. The total number of independent spectral channels is therefore $\mathcal{N}(H) = \ln(\omega_{\rm UV}/\omega_{\rm IR}) = \ln(c/\epsilon H)$. Since the total gravitating vacuum energy is fixed at the classical scale $\rho \sim H^2/G$ (§7.2) and the trace-out compression noise (§7) is distributed equally across these channels, each channel carries a fraction $1/\mathcal{N}$ of the total, so $\rho_\Lambda \propto 1/\mathcal{N}$. This is a non-perturbative result: standard QFT gives corrections suppressed by $(H/M_{\rm Pl})^2 \sim 10^{-122}$, whereas the spectral response gives $1/\mathcal{N} \sim 1/140$. Normalizing at H_0 and Taylor-expanding:

$$\boxed{\nu_{\rm OI} = \frac{\Omega_\Lambda}{2\,\ln(c/\epsilon H_0)}}$$

With $\epsilon = 2\,l_p$, $H_0 = 67.4$ km/s/Mpc, and $\Omega_\Lambda = 0.685$: $\mathcal{N}(H_0) \approx 139.6$, giving $\nu_{\rm OI} = (2.45\pm0.03)\times10^{-3}$. The result is insensitive to H_0 (which enters only inside the logarithm) and yields an independently testable ratio: $\nu_{\rm OI}/\Omega_\Lambda = 1/(2\,\ln(c/\epsilon H_0)) \approx 0.00358$.

DESI's 2024–2025 data releases [22, 30] report evidence for evolving dark energy at 2.8σ–4.2σ. Direct RVM fits by de Cruz Pérez, Gómez-Valent, and Solà Peracaula [31] find $\nu \sim \mathcal{O}(10^{-3})$ with $\nu = 0$ disfavored at 2.7σ–3.1σ. The framework's prediction lies within the phenomenological range from all major dataset combinations. The outstanding discriminating tests are whether the one-parameter RVM form fits the data as well as or better than the two-parameter $w_0 w_a$ model, and whether the measured ν/Ω_Λ ratio converges to the predicted value.

8.2 Gravitational Wave Echoes

A black hole horizon represents a regime where the causal partition differs from the cosmological one. For an observer near the horizon, degrees of freedom that are part of the cosmological visible sector become locally inaccessible — reclassified from visible to hidden sector as they cross the horizon. The effective partition shifts, and the emergent description must shift accordingly.

At a proper distance $\sim \epsilon$ outside a black hole horizon r_h, the local wavelength of an infalling mode reaches ϵ — the discreteness floor. The mode has no kinematically accessible states at shorter wavelengths and must scatter back. The predicted time delay is:

$$\Delta t_{\rm echo} \approx \frac{r_h}{c} \ln\left(\frac{r_h}{2\,l_p}\right)$$

For a $30M_\odot$ remnant with $\epsilon = 2\,l_p$, the logarithmic dependence on ϵ makes this prediction insensitive to the exact value: replacing l_p with $2\,l_p$ shifts the delay by $\ln 2 \cdot r_h/c \approx 0.1$

ms, well within the ~ 54 ms timescale [23]. The exact reflection amplitude requires non-perturbative calculation. The echo signal, if detected, would constitute direct evidence that the discreteness scale ϵ identified by the self-consistency argument (§6) has observable consequences at horizons other than the cosmological one — a regime where the partition-relativity of the emergent description (§9.2) becomes empirically accessible.

8.3 Conjunction as Discriminant

The conjunction of confirmed dark energy evolution *and* detected GW echoes would uniquely favor an information-theoretic origin over standard QFT-derived RVMs, which predict $\Lambda(H)$ running but have no mechanism for discreteness-scale echoes. Each prediction alone is compatible with other frameworks; their joint confirmation would be distinctive.

9. DISCUSSION

9.1 Interpretive Consequences

In the cosmological application, the hidden sector is the trans-horizon region; the degrees of freedom involved in quantum experiments — photons, electrons, slits, detectors — are all visible-sector objects. Their quantum behavior is not produced by a separate marginalization over nearby modes but is a downstream consequence of the single cosmological trace-out: the emergent quantum mechanics, once established by the theorem, governs all visible-sector dynamics, including the local experiments discussed here.

The theorem resolves several foundational puzzles as direct consequences rather than additional postulates.

In the double-slit experiment, the particle traverses a single slit at the level of the deterministic substratum — it has a definite trajectory in the total phase space. The interference pattern arises because the observer does not have access to the total phase space. The transition matrix $T_{ij}(t_2, t_1)$ encodes the boundary conditions of the full experimental geometry, including the number of open slits. Opening or closing the second slit changes those boundary conditions, which changes the transition probabilities and hence the distribution of detection events. What appears as "wave-like" behavior is the trace-out-induced dynamics responding to a change in the visible-sector geometry. The which-path story follows the same logic: a detector at one slit couples the particle's trajectory to additional visible-sector degrees of freedom, altering the transition matrix and eliminating the interference terms.

In Wigner's friend scenarios, the Friend's measurement produces a definite outcome: the indivisible dynamics of the total system (Friend + measured system + hidden sector) evolve to a single state. Wigner, who lacks access to the lab's internal degrees of freedom, must trace them out — producing a superposition assignment that reflects his epistemic deficit, not the Friend's physical state. There is no contradiction because the two descriptions correspond to different partitions, and partition-relativity (§1.4) permits different partitions to yield different reduced descriptions.

The Everettian measure problem dissolves because the total system evolves deterministically as a single reality — the Schrödinger equation is a mandatory compression algorithm

for an observer who cannot track hidden-sector degrees of freedom, and "branches" are features of the compressed description, not of the underlying dynamics. The question "why does the Born rule give the right probabilities across branches?" is replaced by the answer that the Born rule *is* the equilibrium distribution of the indivisible stochastic process, derived rather than postulated.

9.2 Scope of the Theorem

The characterization theorem (§3.3) establishes a full equivalence: a stochastic process on a finite configuration space is quantum mechanics if and only if it arises from embedded observation under conditions (C1)–(C3). The theorem does not identify which physical systems satisfy the conditions (that is empirical) nor derive $\hbar$ for any specific system (that requires Part II). Different partitions produce different reduced descriptions — the equivalence says "quantum mechanics and embedded observation under (C1)–(C3) are the same thing," not that quantum mechanics is the unique description of reality. In our universe, where null geodesics define the causal partition and all sub-c observers share the same horizon, the emergent description is universal.

9.3 Dependence on Barandes' Correspondence

The theorem rests on Barandes' stochastic-quantum correspondence [10, 11] as its mathematical engine. The correspondence is bidirectional and exact, proved via the Stinespring dilation theorem and standard linear algebra, published in a peer-reviewed venue [11], and independently corroborated by Calvo [12] (infinite-dimensional extension), Pimenta [25] (geometric characterization), and Doukas [26] (structural properties). The dependence is load-bearing but circumscribed: partition-relativity, emergent stochasticity, and P-indivisibility (§§1.4, 2) do not invoke Barandes at all. The correspondence enters at a single step: the map from P-indivisible process to quantum mechanics (§3.1).

The correspondence is a 2023 result that has not yet undergone decades of scrutiny. A restriction that narrowed the class of admissible processes could affect the theorem's scope, though this risk is mitigated by the fact that the cosmological trace-out produces a highly structured process whose hypotheses are satisfied directly.

9.4 The Status of the Discreteness Scale

The discreteness scale ϵ may appear to smuggle in a quantum assumption. It does not. Axiom 2 requires a finite-dimensional configuration space — motivated on purely classical grounds by the fact that a finite-area boundary admits a finite number of coupled modes. The result $\epsilon = 2\,l_p$ is a *consequence* of the self-consistency loop (§6), not an input. The Planck length itself is a combination of c, G, and $\hbar$ — but here $\hbar$ is derived (§5.2), so l_p is determined by the same loop. No quantum concept is presupposed.

The deeper question is whether any framework claiming to derive quantum mechanics must posit *some* form of discreteness to avoid an infinite-dimensional configuration space — in which case Axiom 2 would be a necessary structural ingredient rather than an ad hoc addition. Holographic entropy bounds [27] provide independent, non-quantum motivation: the maximum entropy of a region bounded by area A is $A/(4\,l_p^2)$, suggesting that finite-area boundaries enforce finite-dimensionality regardless of the theory used to describe the interior.

9.5 Observable Vacuum Energy Effects

If the quantum vacuum energy is compression noise, why do effects like the Casimir force and the Lamb shift produce observable consequences? Because these are predictions *within* the emergent quantum description, which is empirically exact for all visible-sector physics. The Casimir force depends on energy *differences* between boundary configurations; the Lamb shift depends on virtual photon fluctuations coupling to atomic levels. Both are relative effects within the quantum description. What the framework denies is that the *absolute* zero-point sum appears as a gravitational source, because gravity operates at the classical level that is logically prior to the quantum description (§7.2).

9.6 Relation to Prior Work

't Hooft's cellular automaton interpretation [1] is the closest predecessor, proposing quantum mechanics emerges from deterministic dynamics through information loss. The present framework differs in mechanism (epistemic trace-out vs. microscopic information loss), generality (any system satisfying the axioms vs. particular dynamical rules), and Bell's theorem (outcome independence violation vs. superdeterminism).

QBism and relational quantum mechanics agree that the quantum state is observer-relative and epistemic, but the present framework provides a structural account of *why* (the trace-out applied to the causal partition) and makes quantitative predictions that purely interpretive programs do not.

Jacobson's thermodynamic derivation [16] derives gravity from thermodynamics applied to local horizons; the present framework inverts the direction, deriving quantum mechanics from the causal structure that gravity creates. The two are complementary, and their consistency is a non-trivial check on both.

AdS/CFT. The boundary CFT's completeness follows because the boundary observer is *external* to the system, immune to the inferential limits constraining embedded observers [19]. The holographic description works precisely because no causal partition restricts the boundary observer's access.

9.7 Assumptions, Limitations, and Falsifiability

The theorem requires Axiom 2 (independent assumption in Part I; consequence of the partition geometry in Part II) and Barandes' correspondence (entering at a single step, §3.1). The genericity assumptions — non-degenerate spectrum and non-vanishing configuration-basis overlaps — hold for all but a measure-zero set of systems. The tensor product structure of the emergent QFT follows from spatial locality and is standard for lattice systems; a rigorous proof that Barandes' correspondence preserves spatial locality is expected but not provided here (required for the CC dissolution but not for the derivation of $\hbar$).

The Planck-scale ordering. The framework's ontological commitment — classical spacetime prior to quantum mechanics — may break down at the Planck scale. The framework does not address the deep Planck regime; its domain is the macroscopic causal structure of the cosmological horizon. The self-consistency loop fixes ϵ at $2\,l_p$, above the scale where the ordering might break down.

Falsifiability. The *theorem* would be falsified by a restriction of Barandes' correspondence excluding the processes generated here. The *application* would be falsified by the definitive absence of dark energy evolution or gravitational wave echoes at the predicted timescale, while leaving the theorem intact.

10. CONCLUSION

Two results at different levels of generality have been established.

The general theorem (Part I). Under four axioms (deterministic dynamics, finiteness, a causal partition, and classical probability), the P-indivisibility of the reduced dynamics is proved as a mathematical theorem: any bijective dynamics on a finite system with non-trivial coupling (condition C1) necessarily produces P-indivisible stochastic dynamics in the visible sector. Conditions (C2) and (C3) — inverted timescale separation and sufficient capacity — ensure the physical persistence and observational relevance of this indivisibility. By Barandes' stochastic-quantum correspondence, the embedded observer's reduced description is necessarily quantum mechanics. Crucially, the continuous-time transition data rigorously locks the quantum phases, resolving the Schur-Hadamard gauge freedom and leaving no physical ambiguity in the emergent unitary evolution. The Schrödinger equation, the Born rule, and Bell inequality violations are structural consequences. The action scale $\hbar$ is uniquely determined by the local structural properties of the partition boundary. No quantum postulates are required. The characterization theorem (§3.3) establishes that the conditions are not merely sufficient but *necessary*: any P-indivisible process, realized through a deterministic dilation, requires non-trivial coupling (C1), slow-bath memory (C2), and sufficient hidden-sector capacity (C3). Quantum mechanics and embedded observation under (C1)–(C3) are equivalent.

The cosmological application (Part II). The cosmological horizon provides a physical realization where the sharp-partition approximation is exact and all three conditions are satisfied. The theorem yields: (a) the exact value $\hbar = c^3\epsilon^2/(4G)$ from partition-relativity, the structural/volumetric distinction, dimensional analysis, and thermal self-consistency, with $\epsilon = 2\,l_p$ and the Bekenstein-Hawking entropy formula $S = A/(4\,l_p^2)$ recovered as consequences; (b) a fundamental dissolution of the cosmological constant problem — contingent on the emergent QFT inheriting spatial locality from the classical substratum (§7.1) — recognizing the 10^{122} discrepancy not as a catastrophic fine-tuning failure, but precisely as S_{dS} — the information compression ratio of the observer's quantum description; and (c) falsifiable predictions including dark energy evolution in RVM form with a computed coefficient $\nu_{\mathrm{OI}} = \Omega_\Lambda/(2\,\ln(c/\epsilon H_0)) = (2.45 \pm 0.03) \times 10^{-3}$, consistent with current observational fits, and gravitational wave echoes.

The universe is not broken. The apparent conflict between its two most successful theories is the information-theoretic cost of observing it from within.

DECLARATION OF AI-ASSISTED TECHNOLOGIES

During the preparation of this work, the author used **Claude Opus 4.6 (Anthropic)** and **Gemini 3.1 Pro (Google)** to assist in drafting, refining argumentation, and verifying bibliographic details. The author reviewed and edited the content and takes full responsibility for the publication.

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